

AMSTERDAM SCHIPHOL · FLIGHT OPERATIONS ANALYSIS

Inside Schiphol's Delays

What one month of arrival & departure data reveals about when, where, and why flights run late

42,494

operating flights

~1 month

of arrivals & departures

5 datasets

Schiphol API + 4 lookups

2 models

linear & logistic

Who this is for and what we're asking

TARGET AUDIENCE

- Schiphol airport operations management — staffing, gate & terminal planning.
- Airline station managers operating flights at the hub (esp. KLM).

THE QUESTION

- When and where is the airport most strained, and who is most exposed to delay?
- Can basic operational data predict whether a flight runs late?

THE DATA

Schiphol public API

Anchor dataset: ~1 month of arriving and departing flights — flight numbers, scheduled & actual times, terminals, gates, aircraft, routes, codeshares.

Pulled in a separate notebook (the API crashed when pulling and analyzing together) and read in as CSV.

+ FOUR OUTSIDE LOOKUP DATASETS

- Airport, country & continent reference (route geography)
- OpenFlights airline codes (decode IATA prefixes)
- OpenFlights aircraft codes (readable aircraft names)

Cleaning & feature engineering

~150,000

raw API rows (codeshares counted separately)



42,494

operating flights after de-duplicating codeshares

≈ 50 / 50

split of arrivals vs departures

KEY ENGINEERED FEATURES

delay_minutes & status

Actual vs scheduled time; flagged on-time / early / delayed using the IATA 15-minute threshold.

Time-of-day features

Hour, day-of-week, and grouped periods (Morning, Afternoon, Evening, Overnight).

Cleaned categoricals

Codeshare frequency, integer terminals, readable service types & aircraft names; cargo → 'Cargo' terminal.

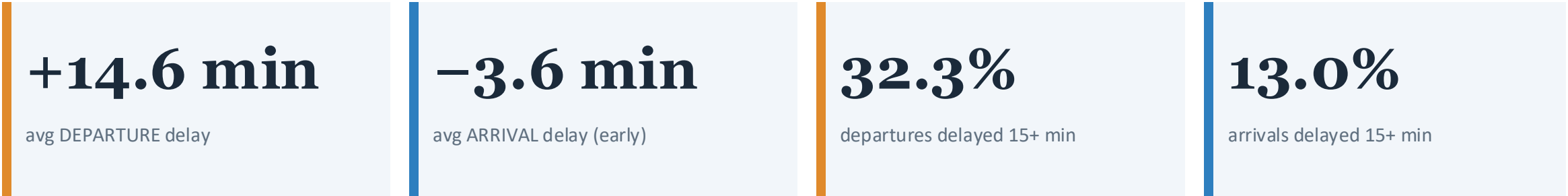
Model dummies

is_klm, is_peak_hour, is_weekend, top-10 airline group, plus one-hot route/terminal/time.

Exploratory Data Analysis

When the airport is busy · who flies it · where flights go · when delays strike

Departures are the delay problem, not arrivals

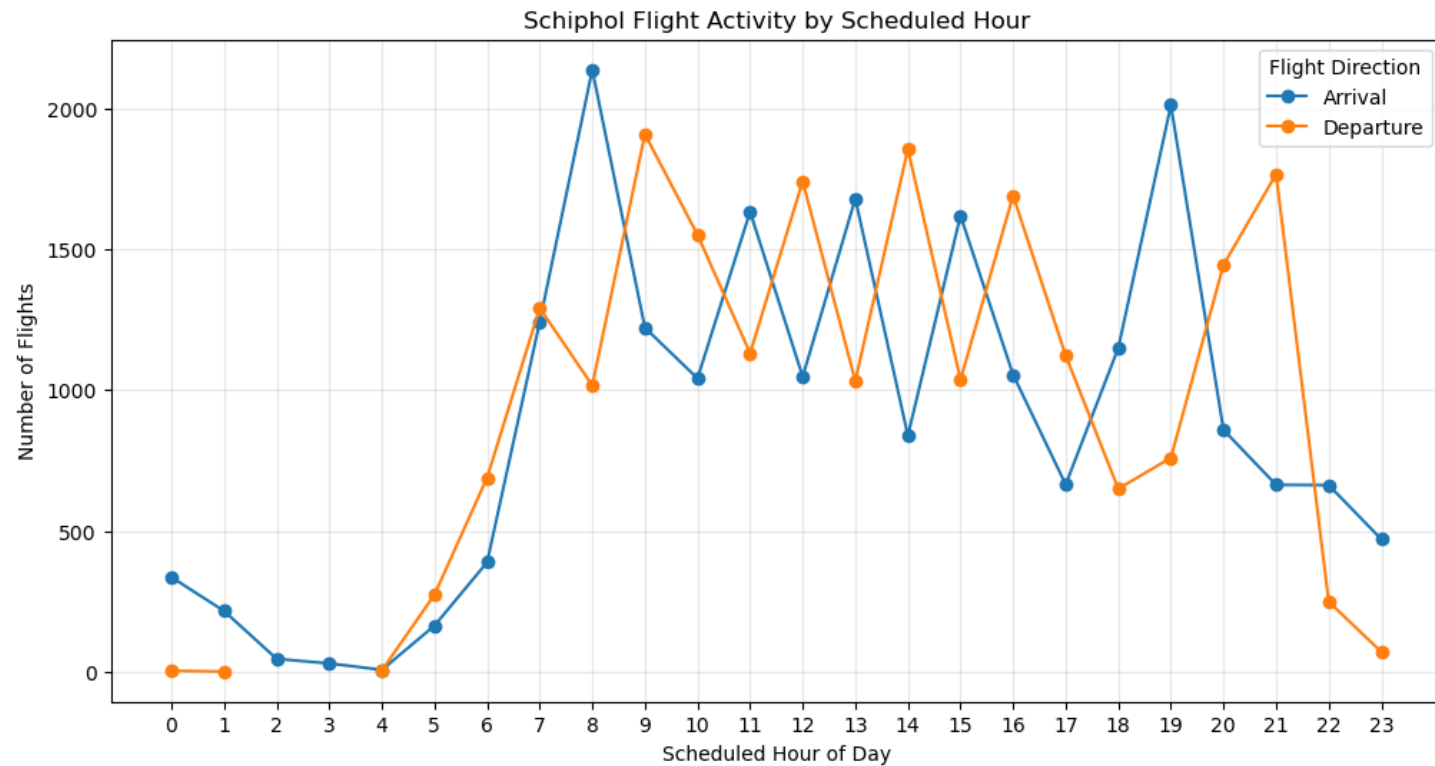


By direction	Flights	Avg delay (min)	Median (min)	Delayed 15+
Arrivals	21,192	-3.64	-9.0	13.0%
Departures	21,298	+14.57	+8.0	32.3%

WHAT IT MEANS

- Volume is balanced — ~21k each way — but delay risk is not.
- The average departing flight clears the IATA 15-minute delay line; the average arrival lands early.
- Outbound operations are where Schiphol should focus delay-reduction effort.

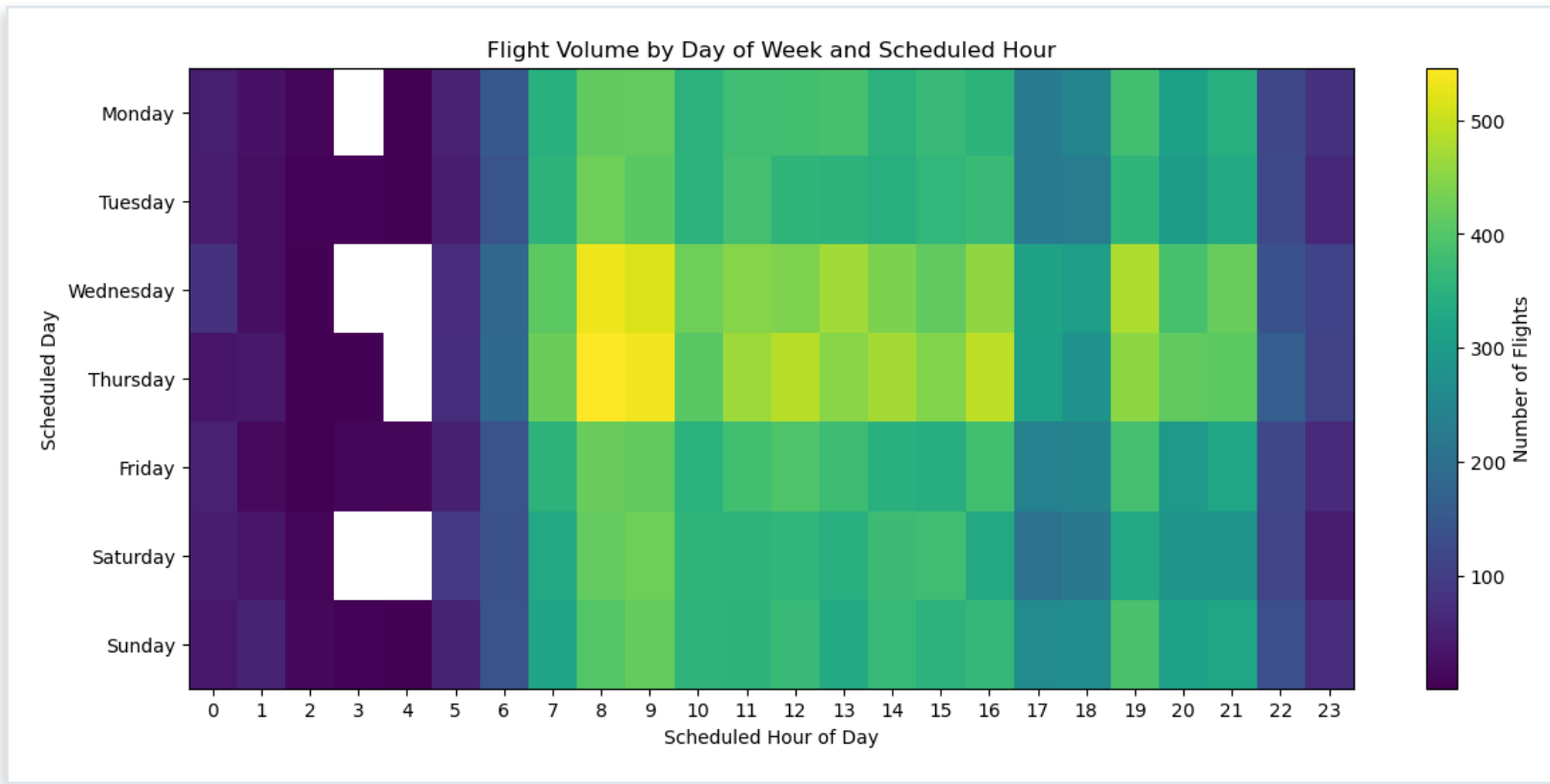
Activity comes in arrival & departure waves



READ THE RHYTHM

- Almost no traffic 02:00–05:00 — then a sharp morning ramp.
- Arrivals and departures alternate in waves: banks of inbound flights feed banks of outbound connections.
- Sustained pressure from ~07:00 to ~21:00 — the window where resources are most constrained.

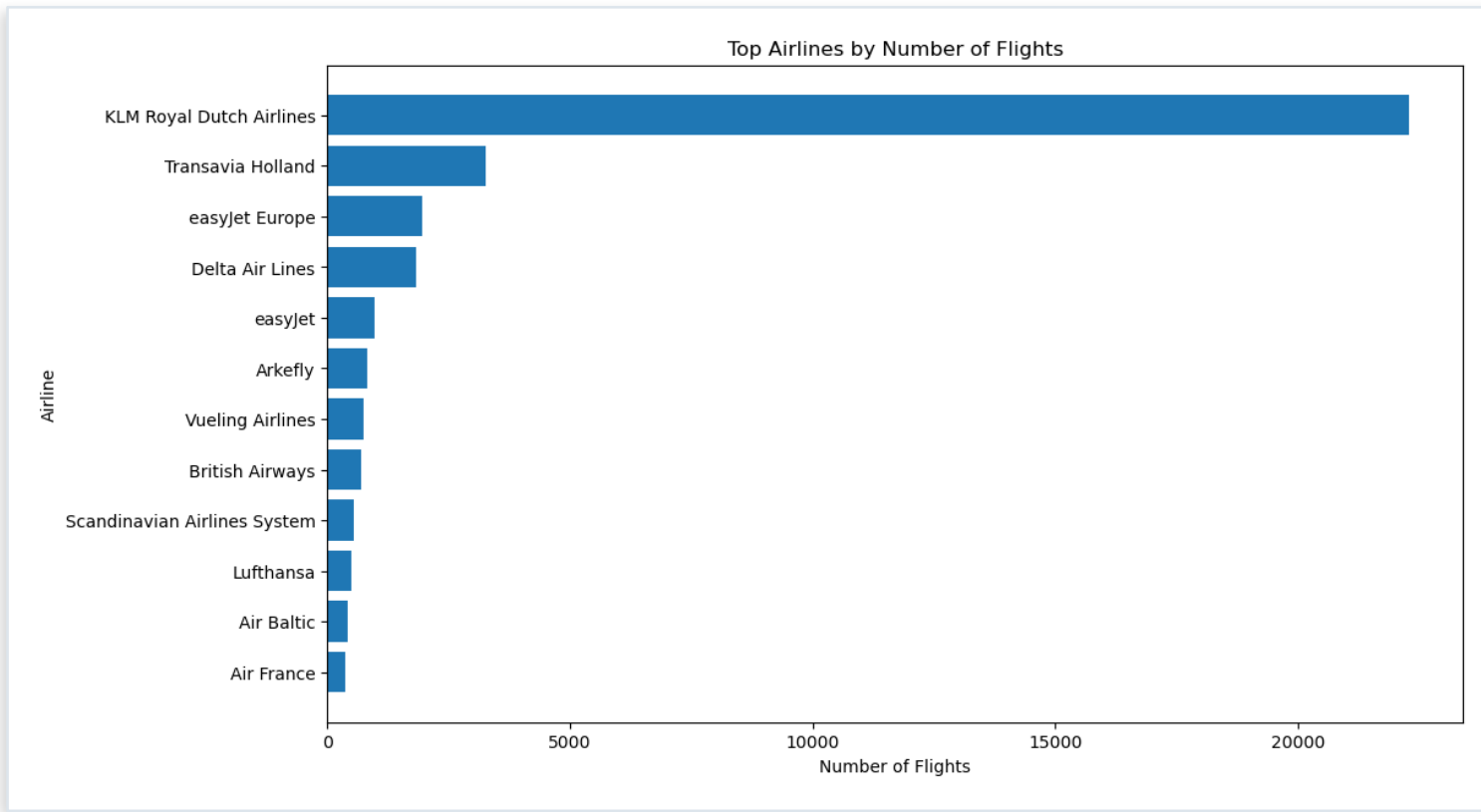
Wednesday & Thursday are the busiest days



READ THE HEATMAP

- Wed & Thu carry ~3,500+ flights each way — clearly above the other days.
- The hot band runs 06:00–21:00 every day; nights are nearly empty.
- Predictable peaks make staffing and gate planning a tractable problem.

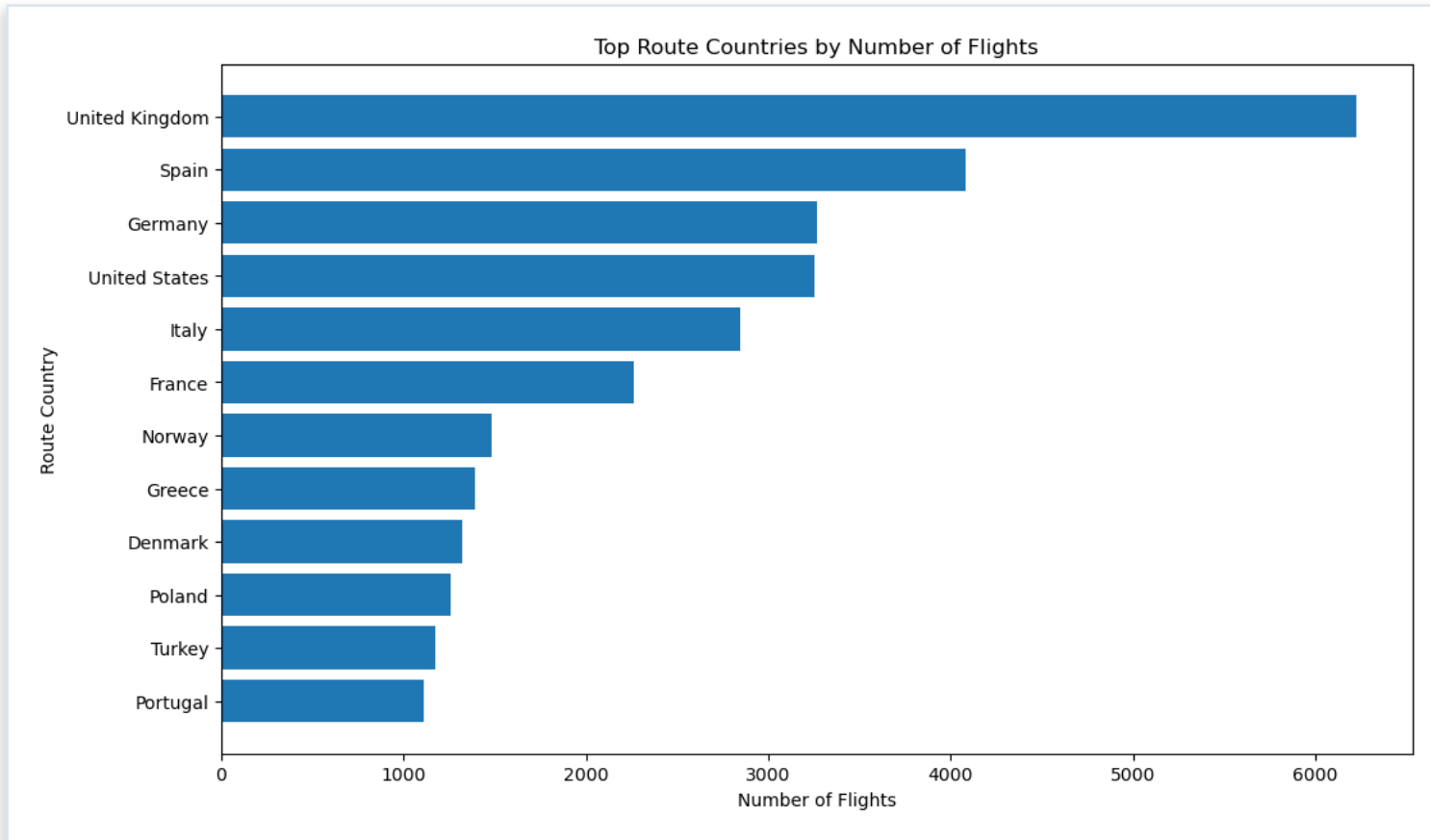
One carrier dominates the hub: KLM



CONCENTRATION

- KLM operates 22,293 flights — over half of all traffic in the sample.
- Transavia (3,265), easyJet Europe, and Delta follow well behind.
- Hub concentration means KLM's operational performance largely is Schiphol's performance.

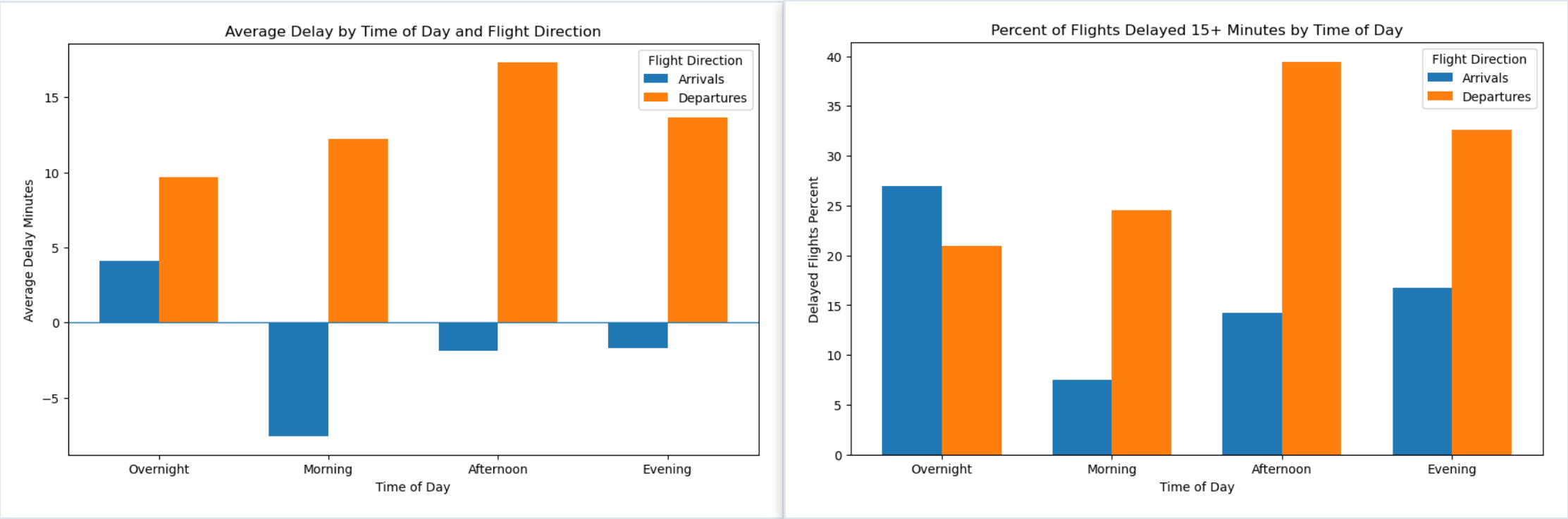
A European network with one long-haul standout



ROUTE GEOGRAPHY

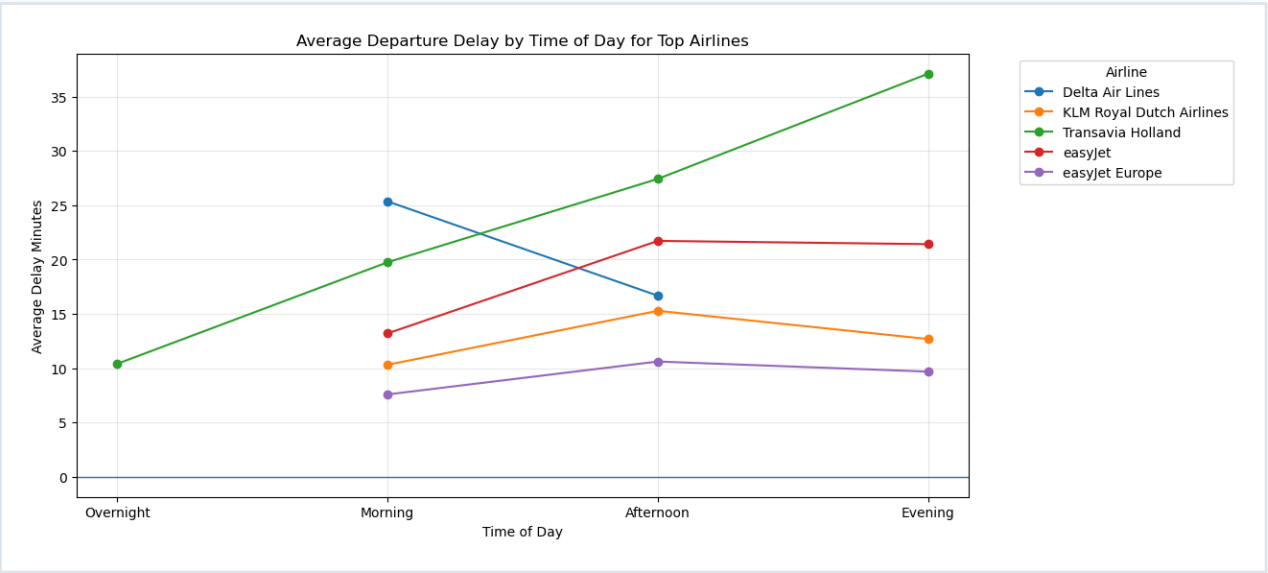
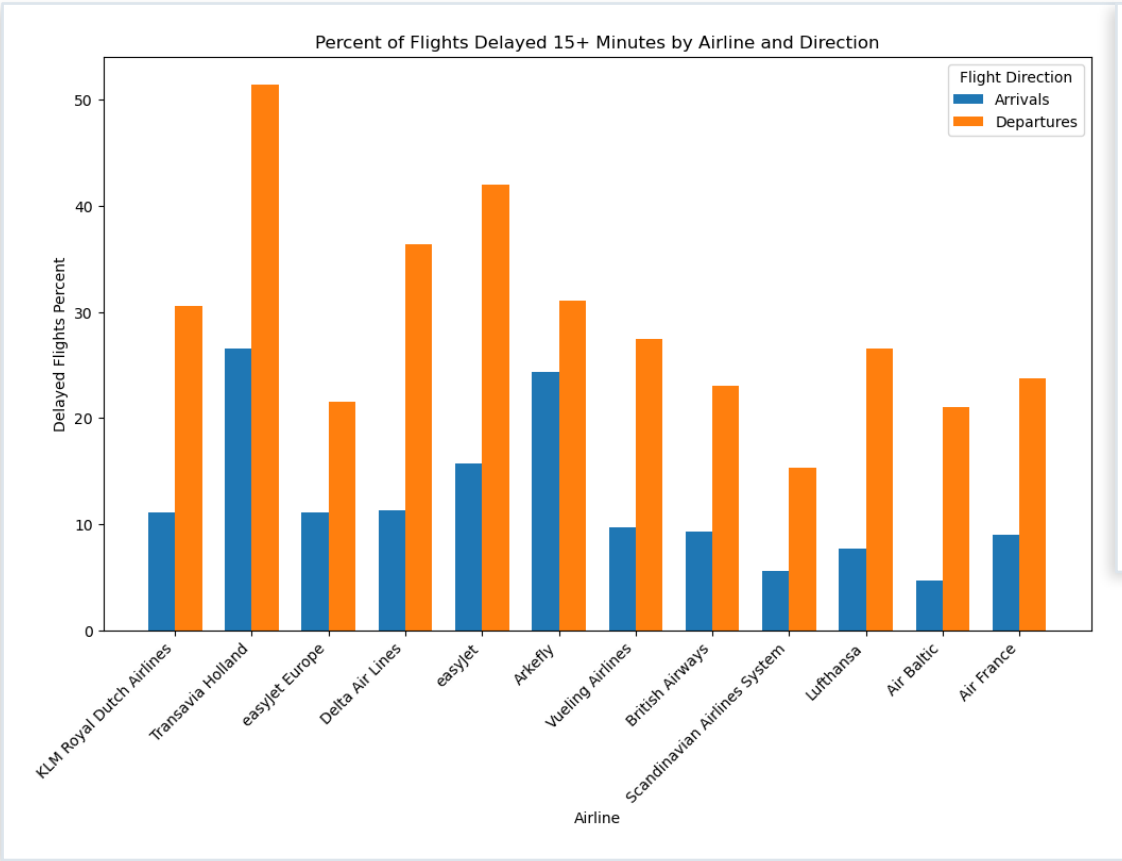
- Top countries: UK (6,221), Spain, Germany, and — surprisingly — the United States (3,253).
- The US is the lone long-haul market in the top 5: Delta–KLM joint venture + connecting-hub role.
- Long-haul departures (Asia, Africa, North America) carry higher average delays than European ones.

Afternoon departures are the worst



Afternoon departures: +17.3 min average and **39.5% delayed 15+ minutes** — the single riskiest slot. Arrivals stay early all day; departure delay builds as the day's disruptions accumulate. If Schiphol targets one window, it's the afternoon outbound push.

Delay exposure varies by airline & deepens into the day

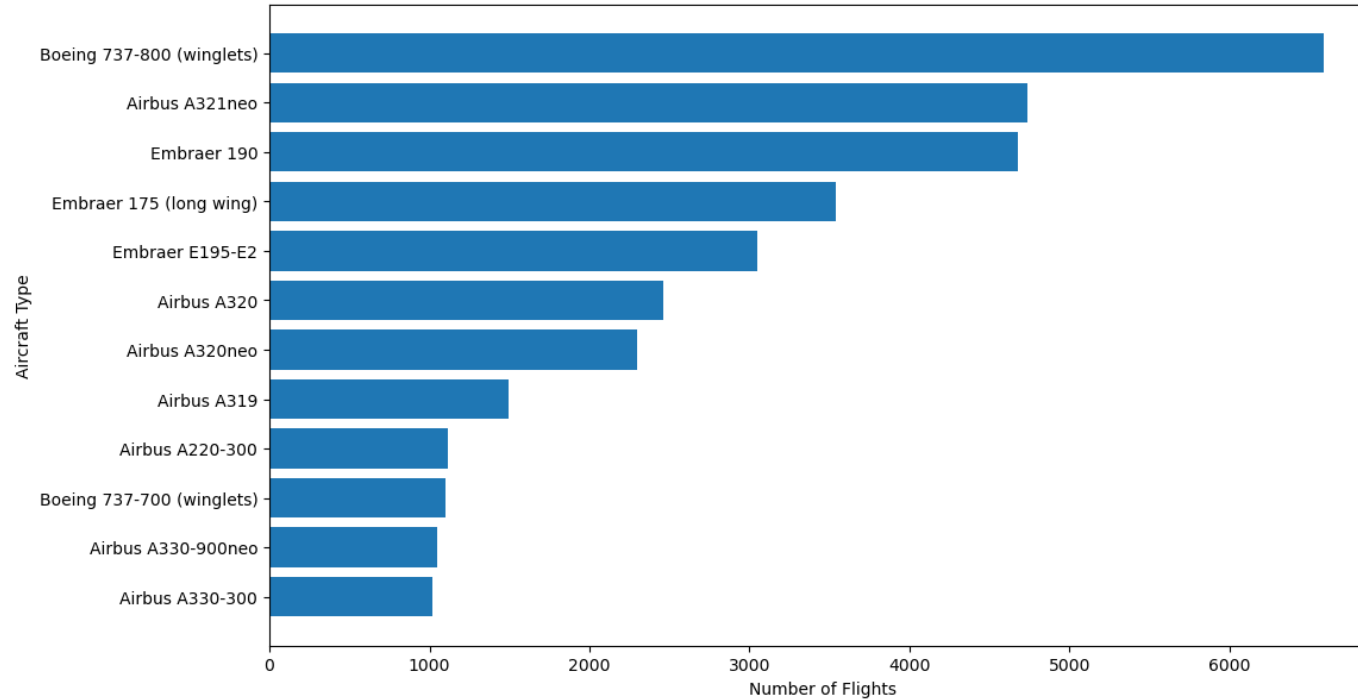


PATTERN

- Departure delay rises steadily from morning to evening for the top carriers.
- Transavia's outbound delay climbs sharply by evening; KLM stays comparatively flat despite its volume.

What flies, and where the gates fill up

Top Aircraft Types by Number of Flights



FLEET MIX

- Narrow-bodies dominate: Boeing 737-800 (6,586) and Airbus A321neo (4,734) lead.
- Embraer regional jets round out the top of the list — short-haul European feed.

TERMINALS

- Terminal 1 handles the most departures (~11k) and shows the highest delay share among the named terminals.
- 'Unknown'-terminal flights have abnormally high delays — a data-quality flag, revisited in the model.

Predicting Delays

Two models: estimate delay minutes, then classify the 15-minute delay

Predicting delay minutes

SETUP

- Target: delay_minutes. Features: direction, hour, time-of-day, day, airline group, route continent, terminal, codeshare frequency + engineered flags.
- 80/20 train-test split; categoricals one-hot encoded.

MODEL vs. BASELINE

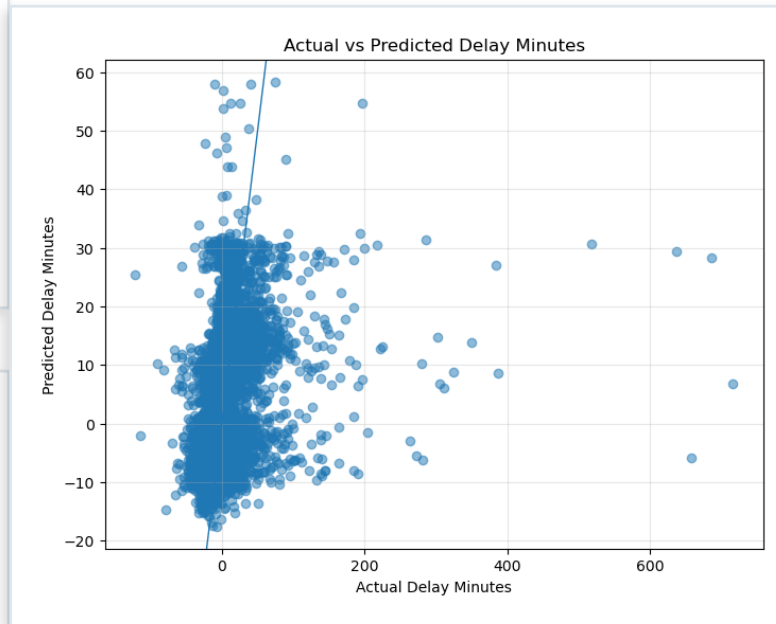
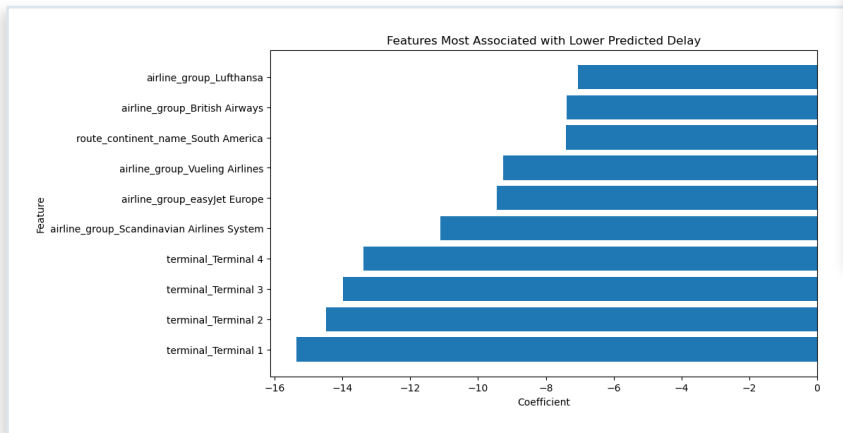
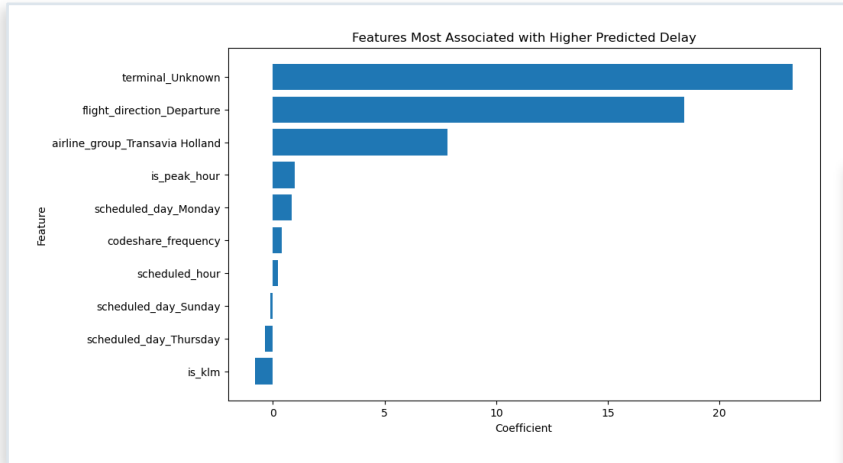
Metric	Baseline (mean)	Linear model
MAE	17.32	15.47
RMSE	32.11	30.26
R ²	0.000	0.112

$$R^2 = 0.112$$

The model explains ~11% of delay-minute variation — and that's expected.

- It clearly beats a mean-guess baseline (lower MAE/RMSE, R² above 0), so it is learning real signal.
- But delays are driven largely by factors absent from the API — weather, ATC restrictions, inbound aircraft, maintenance, crew.
- Long, legitimate outlier delays (multi-hour) were kept, which caps R² but reflects reality.

Drivers of predicted delay



SIGNALS

- Higher delay: departures, peak hours, Transavia, and the 'Unknown' terminal.
- Lower delay: arrivals into the main terminals; several legacy carriers.
- The large 'terminal_Unknown' coefficient tracks records missing actual times — a reporting-quality artifact more than a real driver.
- Actual-vs-predicted scatter: predictions cluster near typical values and can't reach the extreme outliers.

Will a flight be delayed 15+ minutes?

78.1%

accuracy

62.8%

precision

7.1%

recall (at 0.50)

0.709

ROC-AUC

THE RECALL PROBLEM

- At the default 0.50 threshold the model is accurate and precise but far too conservative.
- It catches only 7% of flights that are actually delayed — useless for an early-warning tool.
- ROC-AUC of 0.71 shows the model can separate delayed from on-time flights; the threshold is the lever.

WHY RECALL MATTERS HERE

For airport operations, a missed delayed flight is costly. A flag that fires too rarely defeats the purpose — so we'd rather tolerate some false alarms to catch more true delays. That means lowering the decision threshold.

Trading precision for recall

Threshold	Accuracy	Precision	Recall	F1
0.50 (default)	0.781	0.628	0.071	0.128
0.30	0.704	0.385	0.529	0.446
0.25 ← recommended	0.666	0.362	0.632	0.460
0.15	0.533	0.303	0.827	0.444
0.10	0.370	0.257	0.953	0.405

OPERATIONAL READ

- Lower threshold = bigger net: recall climbs as precision falls.
- At 0.25–0.30 the model catches 53–63% of true delays while keeping accuracy near 0.70.
- An operations manager would likely choose ~0.25–0.30 — a workable balance for an early-warning flag.
- F1 peaks around this range too, confirming it's the sweet spot for this feature set.

CONCLUSION

Schiphol's delay risk lives in afternoon departures

WHAT WE FOUND

- Volume is balanced, but departures drive delay: +14.6 min avg and 32% delayed vs. early arrivals.
- Activity peaks Wed/Thu and across the 06:00–21:00 band; afternoon outbound is the riskiest slot (39.5% delayed).
- KLM dominates the hub; long-haul departures carry higher delay than the European core.
- Models find real but limited signal — useful for ranking risk, not exact prediction, from operational data alone.

WITH MORE TIME

- A full year of data to capture seasonality and stabilize route-level conclusions.
- Add weather, ATC restrictions, runway use, and inbound-aircraft rotation — the real delay drivers.
- Try tree-based models (random forest, gradient boosting) for non-linear delay patterns.
- Fix the 'Unknown' terminal reporting gap to remove a data-quality artifact from the model.